National Institute of Textile Engineering and Research (NITER)

Additional Page No.................... Invigilator's Signature... Tasmin ...Date.......................

Student's ID............................ Registration No .... Course Code......................

Please start writing from below this line

Sheet: Integration Methods.

Q:- 2, 4, 5, 6, 10, 11, 12, 13, 14, 15, 18, 22, 27, 29

H.W :- 6, 13, 23, 25, 33

Sheet: Improper Integral:

Definations: Improper integral, kinds of improper integral.

Q:- 2, 3, 5, 7, 10

H.W:- 2, 3, 5

Sheet:- Numerical integration.

Statement: Trapezoidal, simpson's $\frac{1}{3}$rd and $\frac{3}{8}$th rule

Math:- Using simpson's $\frac{1}{3}$rd and $\frac{3}{8}$th rule.

- $\int_1^2 \frac{1}{x}\,dx$
- $\int_0^\pi \sin x\,dx$ by dividing the interval into eight strips
- $\int_0^2 \sqrt{x(4-x)}\,dx$ with $h = 0.25$ upto 4 decimal places.

Sheet:- Arc Length.

Q:- 3, 5, 6, 8, 9, 11, 12

Sheet:- Area of a surface of Revolution

Q:- 2, 4, 5, 7

Sheet:- Differential Equation:

Definations: ODE, Types of D.E, Order & degree of D.E, LDE, IVP;

Preparing D.E: E-2,3,5,6, 8

Seperation of variable: Ex-2, 4, 5, 7, 8

Homogeneous: Ex-1, 2, 3

Exact: Necessary and sufficient condition (proof)

Ex-3, 4, 6, 7, 9. H.W-1 (Procedure-B), H.W-3 (Procedure-D)

Linear D.E: Find the solution of $\frac{dy}{dx} + Py = Q$.

* State and proof Bernoulli's Equation

Q:- P-1, 3, 5, 6

Homework:- 2, 5, 6

Sheet; Mathematical Modeling with D.E

Q:- P-1,2

H.W:- 1, 3, 5

Sheet; Sequence and Infinite Series

Definations: convergent sequence, Monotonic sequence, series, convergent, Divergent, oscillatory series, Properties of Infinite series, Alternating series, Absolutely & conditionally convergent)

Page-3: E-4: $\sum \frac{2n+1}{n^2(n+1)^2}$ H.W: 6, 7.

Convergent Test:-

Divergent Test	P-series test Proof (statement)	comparison test statement	D'Alemberts statement	Raabe's statement	Leibnitz's Rule statement
E-2 H.W-2	E-1	E-2,3,6 H.W-2,6,7,10,12	E-2 H.W-3	E-1,3,5,6,7 H.W-4	E-2,3 H.W-3

Page-12: E-3 ; H-W-2

Maclawrin Polynomial & Taylor's polynomial.
(E-2, H.W-1)   E-1, E-2 (sigma notation)

Statement: Taylor's formula with remainder

Taylor's series: Q-1, 3   Maclaurin series: Q-4

H.W - 1, 2, 3, 4, 5

<u>Sheet</u>: Numerical solution of ODE:

Taylor's series:- P-1, H.W: (i) $\frac{dy}{dx} = x^2 - y$; $y(0) = 1$, upto $x = 0.2$ with $h = 0.1$

(ii) $\frac{dy}{dx} = x + y$; $y(1) = 0$, upto $x = 1.2$ with $h = 0.1$

(iii) $\frac{dy}{dx} = -xy^2$; $y(0) = 2$ with $h = 0.1$ up to $x = 0.2$

(iv) $\frac{dy}{dx} = e^x - y^2$; $y(0) = 1$ upto $y(0.1)$ correct to four decimal place

Euler's Method:- P-2, H.W: 5, 7, $y' = -2y + x^3 e^{-2x}$, $y(0) = 1$, find (0.5) with $h = 0.1$